Intuition in Design & Emotion?

Transforming raw data into conclusions, a meta-analysis of the 2006 Design & Emotion conference papers

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Abstract

In any research field the transformation of raw data into conclusions is a critical phase, necessitating a systematic approach. The basic question is: How can I as a (design) researcher corroborate my conclusions? This paper is based on a study of the analytical/interpretative portion of papers published in the Proceedings of the 2006 Design & Emotion Conference. We investigate how the authors present the interpretative process in their papers, such as using specific methods for interpretation and analysis. It became obvious that there were few specific methods for interpretation mentioned in the Conference papers studied, though there was a wide range of methodological approaches from theoretical and statistical to purely qualitative. Notably, most papers did not mention any method at all. The issue of interpretation is certainly not new, but in the light of the topic of design for emotions, design (research) *by* emotions or intuition should be discussed.

Conference theme: Methodological issues

Keywords: interpretation, methodology, tacit knowledge

Introduction and aim

One interesting element of the design research process, especially when it comes to the sometimes elusive phenomenon of emotions, is the analysis of the research material. In any research field, the transformation of the empirical material into conclusions is a critical phase, necessitating tools and systematic procedures to ensure validity and reliability of the conclusions. Validity and reliability, in turn, can be argued to be contested measures in qualitative research, but the underlying question still remains: How can I as a design researcher corroborate my conclusions based on the material collected?

> *"To validate is to question, in particular to continually ask what is being investigated and why, and also ask about the when, where and who of an action."* (Kvale, 1989, p.81)

There are various analytical methods and models at hand for the design researcher, some taken from other disciplines, others from within the field. The aim of this paper is to investigate and discuss the ways the interpretation of the research material is reported in the articles studied. This includes the formal presentation of the interpretative portions, as well as methodological issues, such as which methods are explicitly used when interpreting the material. There is also the question of the impact of the personal traits of the individual researchers when it comes to interpreting the research outcomes.

> *"Qualitative research methods are largely subjective in that they rely heavily upon the investigator's skills of observation and interpretation to provide valid information. Therefore, much training and experience is needed to carry out this kind of research."* (Borg and Gall, 1989, p.379)

In order to achieve our goal, the outset of this study needs to be clarified. This is a descriptive attempt, not an interpretative one. In that sense, this paper resembles a review of the articles studied in order to detect patterns in the interpretation phase. Individual papers are not analyzed, though their form and content constitute the raw material of this study; rather it is the collective impression they give that is examined. Our intention is not to pass judgment on the ways interpretations are made in the individual articles. The goal is to raise general questions regarding the manner in which the interpretation of the (raw) research material is described and presented, and to discuss implications for design research. In this sense, this is an explorative venture.

Material and method

Material

This paper is based on a study of the analytical/interpretative portions of papers published in the Proceedings of the Design & Emotion Conference in Gothenburg 2006 (Design & Emotion 2006). The papers (n=82) were retrieved from the Design and Emotion Society webpage.

Method

The methodological approach chosen was based on a series of questions used for evaluating the content and disposition of the papers regarding making the interpretation of the material explicit:

1. Is the article ‘empirical’ or ‘non-empirical’? By ‘empirical’ we mean that it presents original research material. Theoretical discussions or descriptive design cases are classified as non-empirical.
2. If empirical: Is any statistical method used for interpreting the research material?
3. If empirical: Is any other method explicitly mentioned for interpretation; if so, which?
    - If no: Does the article explicitly present an interpretation of the material?
    - If yes: How is the interpretation presented?

Some additional questions were asked when reviewing the articles:

4. If empirical: Are the words ‘reliability’ and/or ‘validity’ mentioned?
5. If empirical: Is the position of the researcher(s) stated? By position we mean experience, competence, knowledge, education, cultural background, or any other aspect of the individual which could affect the process of interpretation.

This set of questions was the starting point of the investigation and could be expanded by a more in-depth qualitative study of the integration of the interpretative process in the texts. In each article the above points were investigated.

Results

Some papers reported several aims and methods. Many reported a series of iterative steps leading to a range of different kinds of empirical data. This made it difficult for the reader to understand if a method used for *categorizing* data also affected the *interpretation* of the data. For these reasons, detailed categorization of the papers in this study was considered difficult and unproductive for the purpose of this review. The findings of this study are presented with a range of examples from the papers (n=82). All articles are not specifically commented on, but are included in the investigation.

25 of the 82 articles were considered to be non-empirical (e.g. Wood; Candy, Edmundson; Kong; Anusas). Of the ones considered empirical, 12 used a statistical method of some sort to analyze all or parts of the empirical material (sometimes in addition to other methods) (e.g. Grimsaeth, Baggerud; Stilma; Schifferstein).

In the empirical papers, there were examples of methods and tools for analysis/interpretation such as text coding (Eliot) or qualitative computer software (e.g. Mason, Porter). These were applied to parts of the empirical material in the respective articles. In a few papers (e.g. Fang, Lin), objective measurements were reported. In a handful, methods for analysis were mentioned but not further discussed (e.g. Kim, Zimmerman). Several papers did not mention any method used for interpreting the empirical material (e.g. Ionascu; Tzvetanova, Tang, Justice; Rodríguez Ramírez; Wikström).

Research tools which provided instant information about the experiences of the participants partly freed the researcher from interpreting the material, such as VAS (e.g. Olander, Sperling), Likert scale (e.g. Hårleman, Werner, Billger), User Compass Chart (e.g. Sperling et al.), Sensagram (e.g. Fenech, Borg) and close-ended questionnaires (e.g. Harrington, Lechner, Simonoff), among others.

We have also reflected upon issues which possibly could influence the interpretation process in each study. The first was if the researchers' position was mentioned in the paper, such as personal experience (e.g. Kim, Zimmerman), knowledge (e.g. Mason, Porter), profession (e.g. Skogen; Gonzalez Veron, Evenson), education (e.g. Lilleng, Baggerud), fundings (e.g. Hakatie, Ryynänen; Chhibber et al.; Sperling et al.) or as a biography (e.g. Huang, Deng).

The second issue was if validity and/or reliability was used or discussed. Just a few studies mentioned validity; sometimes the terms were used without declaring the use or definition of the term, e.g. in terms of repeatability or generalizability (e.g. Laurans, Desmet; Lechner, Harrington, Simonoff).

Interpretation of the results

A main concern of this review was to avoid interpretation of the content of the papers studied. This proved to be quite a challenge, since the structure and disposition of most of them demanded detailed reading to clarify the use of methods, theory, and process of interpretation/analysis.

It became obvious that there were few specific methods for interpretation mentioned in the Conference papers studied, though there was a wide range of methodological approaches from theoretical and statistical to purely qualitative. Notably, most papers did not mention any

method at all. This makes it almost impossible for an outsider (or insider for that matter) to get a feeling for how similar empirical material is handled in this field. Some use statistical methods to analyze results from Visual Analogue Scales, others discuss them more freely.

So, how come some researchers choose to interpret their data in a structured way using well-known tools, whereas most choose a non-explicit method? Several possibilities can serve as explanations. We want to emphasize that the following list should be regarded as speculative and incomplete suggested explanations based on reading the articles included in this study. We have not contacted any of the authors to ask their reasons for not explicitly describing this process.

1. The process of interpreting research material in this community of practice (research in design and emotion) is common knowledge and does not have to be made explicit: Everyone knows how it is done and there is no need to waste time or effort on explaining the obvious.
2. There is no tradition in the design research community of including this aspect of the process in conference papers.
3. It is a practical matter of number of words. The manuscript length limitations (this year 4000 words, including abstract and references) make it difficult to include elements not considered critical.
4. There is a lack of knowledge of how to systematically analyze/interpret raw material or how to describe it in research articles.
5. The authors are not aware of the importance, or even existence, of this phase of the research process.
6. There is a lack of interest in discussing the process of interpretation.
7. The interpretation is made intuitively and cannot be described in text (or it is very difficult and thus omitted).

Again, we want to stress that we have no idea as to the actual reasons for not including a clear presentation of the interpretation of the research material in the articles lacking such. These are just suggestions which can be used as a basis for discussion.

So, how do we counter these explanations, assuming they are true? The first one, that the process of interpretation is shared across the research community and does not have to be explained, is problematic for several reasons. First of all, one can question if this community of practice really is so homogeneous that the outcome of such a qualitative process is transparent and clear to any other member of the community. Looking at the diversity of research approaches and topics among the articles in the Proceedings, along with the backgrounds and

professional competencies of the authors, it is obvious that this is a heterogeneous field. Although a more or less shared basis in the training and education of the researchers may exist, one can still argue that it could be expected that the outcome would vary depending on the individual interpreter. Secondly, isn't the diversity of interpretation one of the strengths of research for design? Sharing the pros and cons of alternative interpretations of the material presented could enrich and vitalize the discussion.

The second explanation, based on a lack of tradition, is a delicate one. Certainly, tradition is of great importance for shaping any branch of research but to answer why and how this tradition has come into existence is difficult. This is not the place to elaborate on design research history and the origin of methodology, nor comment on the format of a particular conference. What is apparent, though, when reading the papers is the many attempts to describe methods and techniques for collecting material, but a lack of similar effort when it comes to evaluation and interpretation. Moreover the reader of the Conference articles has no idea if the author has reflected on the interpretation in the research processes (which is very likely), or not (hopefully unlikely).

The limitations on the number of words for the texts published in proceedings, forces authors to consider what is of greatest interest for the intended audience. If this is the reason why an explicit description of the process of interpretation is omitted, it implies that this is considered less important than an extensive description of the theoretical research background or the design process. Looking at the articles examined, it is obvious that most of the researchers have made this choice, which could be in line with the previous explanation.

The following two possible explanations are not generally considered valid in a research context (lack of knowledge and unawareness). But if this is the reason, it is most certainly in concert with other reasons such as those mentioned in explanations 1, 2, and 7.

A lack of interest in explicitly reporting the interpretation can be a reasonable explanation considering each research project separately, but as a research field, mirrored in the Conference Proceedings, it is arguably more discouraging. There are often lengthy discussions of the design process in the articles, but there is seldom a reflective discussion about the practice of interpretation. So the conclusion could be that it is not considered as important or interesting as

the design process. To be fair, the design process is not discussed reflectively on the most part either in the texts, although it is most certainly practised in the research activity itself.

The last explanation is perhaps the most probable one, although the word ‘intuition’ can be misleading. What we mean by this is that the skills and knowledge accumulated by the researchers during their practice is incorporated to such a degree that in a given situation, appropriate actions and understanding of their effects are almost automated and come naturally without any deeper reflection. In ‘The Reflective Practitioner’, a widely cited and referenced book in the design community, Donald Schön (1987) describes the practice and abilities of a number of practitioners in various fields. Drawing on Polanyi’s notion of tacit knowledge, he claims that over the years, a practitioner develops abilities and strategies which seldom or never are described openly. By reflecting over their ways of dealing with problems within their respective fields, practitioners can improve their own understanding of their practices and thus improve their abilities to handle common as well as unexpected problems they encounter. This reflective practice, hence, is a means of making the tacit explicit. A practitioner, for example an industrial designer, could very well get by without being reflective, and instead rely on gut feeling trial-and-error. We argue that this is not sufficient for a design researcher. In design research, the openness of the ‘how’ and ‘why’ is crucial for disseminating processes and results within the scholarly community.

If interpretation and evaluation of research appears ‘intuitive’ to the reader, in the sense of tacit knowledge, then it is crucial to declare the position and background of the author/researcher since this has direct impact on the interpretation made. It is a matter of transparency and should be made as a courtesy to the readers so they can make up their own minds if the interpretation is within reason. This could be of special interest if there are several authors of an article, especially if their interpretations differ. Sometimes, a conflict of interpretations is more interesting and fruitful than a consensus.

Implications for design research style

Design research, especially in the field of design and emotion – research concerning experiential design such as experiences of pleasure, usability, or emotional responses towards existing products and tentative design proposals – could be said to be based on a hermeneutic tradition. This statement should be taken in its broadest sense, but the common denominator is that this kind of research does not pay homage to the positivistic tradition. There are no (definitive) answers, no correct methods of investigation, and no truths to be discovered. Design research has more in common with the social sciences, such as ethnography, where the position and

experience of the designer plays an important role, both in regard to the research approach and the evaluation (interpretation) of the empirical material produced.

Design research as a field has thus been inspired by the social sciences, and has adapted several methods and research approaches into its own field. One way of establishing design research as an academic field of inquiry is to produce and present relevant and detailed case studies. These are often descriptive in nature, that is to say, the design process, material, methods, and outcomes are presented and discussed. In many cases these different aspects of the research process are presented intermixed in the texts. One thing which is characteristic of design research, and which sets it apart from the social sciences in general, is the attempts to propose future alternatives (or alternative futures) by improving existing ones or proposing novel practices or products. The systematic collection of such successful cases is what makes up science *for* design (Krippendorff, 2006, p. 209).

Doing/making is thus a key feature of research for design. Naturally, this has implications for the interpretation of design research.

- The researcher cannot be objective.
- It is not looking for truths, but propositions.
- The position/experience/knowledge/skill of the researcher/author is important for the process of interpretation.
- The research is conducted in a specific time and place (and culture).
- The designer has some *intentions* (which could be in the form of hypotheses).
- It takes time to create design artefacts used in the research (= less time to interpret?).

Hence, there are many aspects to take into consideration when approaching design research articles.

Conclusions

In this paper we have studied articles published in the Proceeding of the Fifth Design & Emotion Conferences of 2006. Many are descriptive in the sense described above. We have focused on if and how the authors have presented their reflections on the research material presented in each paper.

We have found that the interpretative portion, as an explicit section or separate discussion, is missing in most of the articles. The interpretation is often intertwined with the description of the material and the design process. One explanation is that this is necessary due to the nature of the design research process itself. This approach has several consequences. First it makes it difficult for the reader to evaluate the interpretation of the research. Secondly, it renders a quick overview of large written material almost impossible (as we noticed during our study). Thirdly, it establishes a tradition of devaluating a clear and open discussion on interpretation within this particular research community. For design research to be cumulative, communication and transparency are crucial.

Another consequence of this way of publishing the interpretation of the design research material is that it seems to be partly based on the personal experiences and knowledge of the individual researcher. Combined with a lack of an open and clear reflective presentation of the interpretation of the research outcomes, one risks weakening the credibility of the conclusions. Not presenting alternative and competing interpretations also can imply that the conclusions are present a priori. This may not be the case in any of the articles studied, but as a reader the question is inevitable.

Following this line of reasoning, one might argue for tacit knowledge in design research, in particular in respect to interpretation of the empirical research material. Hence, we want to raise the question of the importance of the personal traits of the individual design researcher when it comes to interpretation. As such, this is not an issue *if* the interpretative process and the conditions for it are made explicit but, if this is missing or unclear, one can wonder what role, if any, intuition plays in research for design and emotion.

We end this discussion with two open-ended (and possibly rhetorical) questions: Can the design research community in general, and qualitative design research in particular benefit from more open and transparent presentations of the interpretation of research presented in conference papers and other design research articles? If so, will this demand a more in-depth declaration of the standpoint, background and knowledge of the researchers?

> *Taken seriously, a science for design is inherently self-reflective, within its boundary, while always open to other stakeholders. It is destined to continually reflect on itself, on its strengths, and on its weaknesses.* (Krippendorff, 2006, p.271)

Finally, we hope that this paper encourages an open discussion on this topic for the benefit of all.

Note

The selection and grouping of the articles are available on request by contacting the authors.

About the authors

Henrik Enquist holds a Master's in Engineering Physics and a Bachelor's in Fine Arts. He is currently a PhD student focusing on how images, technology and visualizations as autobiographic artefacts and processes affect persons in treatment and rehabilitation.

Camilla Nordgren, with an MBA, is currently a PhD student in the field of traumatic spinal cord injury and societal resources. She has a special interest in methodological questions and the consequences of how empirical data are interpreted

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